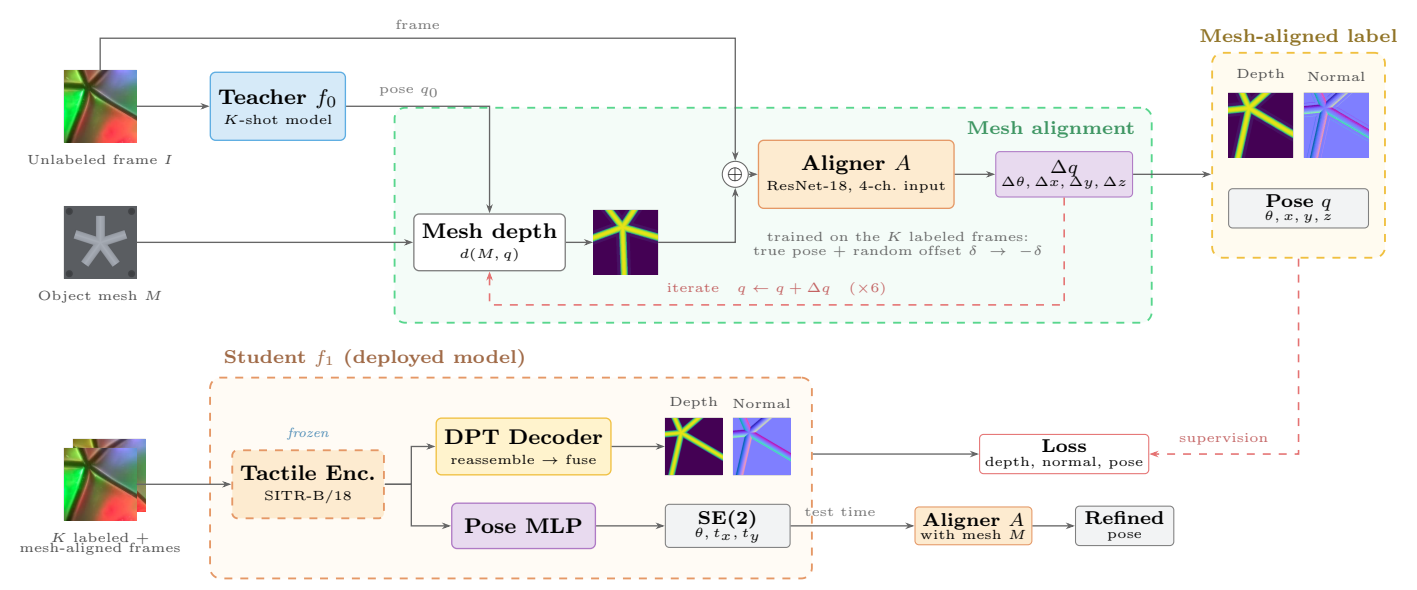

frame
Unlabeled frame I
Teacher f_0
K-shot model
pose q_0
Object mesh M
Mesh depth
$d(M, q)$
Mesh alignment
Aligner A
ResNet-18, 4-ch. input
Δq
$\Delta\theta, \Delta x, \Delta y, \Delta z$
trained on the K labeled frames:
true pose + random offset $\delta \rightarrow -\delta$
iterate $q \leftarrow q + \Delta q$ $(\times 6)$
Mesh-aligned label
Depth
Normal
Pose q
θ, x, y, z
Student f_1 (deployed model)
frozen
Tactile Enc.
SITR-B/18
DPT Decoder
reassemble → fuse
Depth
Normal
Pose MLP
SE(2)
θ, t_x, t_y
Loss
depth, normal, pose
supervision
test time
Aligner A
with mesh M
Refined
pose
K labeled +
mesh-aligned frames